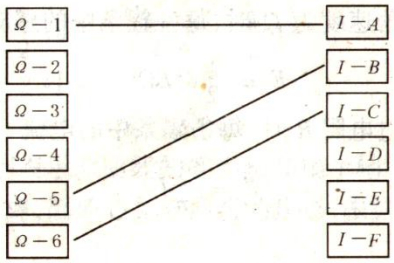
$$\Omega_0 = \frac{4MR}{l^4 B^2} \tag{11}$$

Larger R gives a larger steady angular speed Ω_0, and the steady current I_0 is independent of R.

2.



4.

For charge q, mass m, speed v, if its initial motion is normal to B, a particle in this uniform field feels a Lorentz force; Newton's law gives uniform circular motion. T and R are the period and radius:

$$qvB = m\frac{v^2}{R} = mR\left(\frac{2\pi}{T}\right)^2 \tag{1}$$

Hence:

$$T = \frac{2\pi m}{qB} \tag{2}$$

$$R = \frac{mv}{qB} \tag{3}$$

Here particle **a** initially moves in quadrant I, right of y:

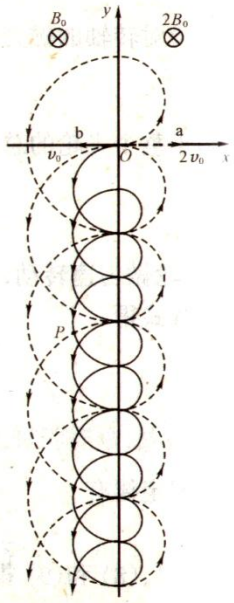


Fig. 20-4-1

$$T_{ar} = \frac{2\pi m}{q(2B_0)} = \frac{\pi m}{qB_0} \tag{4}$$

$$R_{ar} = \frac{m(2v_0)}{q(2B_0)} = \frac{mv_0}{qB_0} \tag{5}$$

After $\frac{1}{2}T_{ar}$, it enters II/III, left of y; its period and radius then become

$$T_{al} = \frac{2\pi m}{qB_0} \tag{6}$$

$$R_{al} = \frac{2mv_0}{qB_0} \tag{7}$$

It repeats the dashed xy path in Fig. 20-4-1.

b too starts in quadrant III, left of y:

$$T_{bl} = \frac{2\pi m}{qB_0} \tag{8}$$

$$R_{bl} = \frac{mv_0}{qB_0} \tag{9}$$

After $\frac{1}{2}T_{bl}$, it enters quadrant IV, right of y; its period and radius then become

$$T_{br} = \frac{\pi m}{qB_0} \tag{10}$$

$$R_{br} = \frac{mv_0}{2qB_0} \tag{11}$$

It repeats the solid xy path in Fig. 20-4-1.

Thus on the right both **a** and **b** have period $T_0 = \pi m/qB_0$, and $2T_0$ on the left.

As Fig. 20-4-1 shows, the paths must cross or touch for a meeting. To find the least delay, let **a** and **b** start together. Compare passage at